ACADEMIC PROJECT REPORT

FACTSHEETAI

A Source-Backed System for Mutual Fund Portfolio Overlap, Concentration and Drift Analysis

Submitted in partial fulfilment of the requirements for the award of

[Degree / Programme]

Submitted by [Student Name] | [Roll Number]

Under the guidance of [Guide Name]

[Department / Institution Name]

Academic Year 2026–2027

Certificate

This is to certify that the project report entitled “FactSheetAI: A Source-Backed System for Mutual Fund Portfolio Overlap, Concentration and Drift Analysis” is a record of project work prepared by [Student Name], bearing [Roll Number], under the guidance of [Guide Name], in partial fulfilment of the requirements of [Degree / Programme] at [Institution Name].

The work presented in this report describes the design, implementation and source-bounded evaluation of a software prototype for analysing disclosed equity holdings in Indian mutual fund portfolios. It is submitted for academic evaluation during the academic year 2026–2027.

Project Guide	Head of Department	External Examiner
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Institution seal: _____

Declaration

I declare that this report, entitled “FactSheetAI: A Source-Backed System for Mutual Fund Portfolio Overlap, Concentration and Drift Analysis,” has been prepared by me as an academic project. The implementation and discussion are based on the project repository, its archived source records, documented methodology and observed application behaviour. Sources used in preparing the report are acknowledged in the references and annexures.

I further declare that the report distinguishes implemented functions from proposed or incomplete functions, and that unavailable source values have not been represented as observed data. The portfolio calculations are presented for educational analysis and do not constitute investment advice.

Place: _____

Date: _____

Student signature: _____

Acknowledgement

I express my sincere gratitude to [Guide Name] for academic guidance, constructive feedback and support during the development of this project. I also thank the faculty members of [Department / Institution Name] for the conceptual foundation and review that helped shape the project's objectives and scope.

The project relies on publicly disclosed portfolio information and scheme records made available through the Association of Mutual Funds in India (AMFI) and asset management company (AMC) disclosures. I acknowledge the providers of these records and the maintainers of the open-source software libraries used in the implementation, including Python, FastAPI, Pydantic, pandas, openpyxl and xlrd.

Finally, I acknowledge the contribution of the project documentation and test fixtures, which record the intended calculations, known limitations and reproducibility controls. All conclusions in this report remain bounded by the available source corpus and the limitations described in the methodology.

Executive Summary

FactSheetAI is a research prototype designed to compare the disclosed equity holdings of Indian mutual fund schemes. It addresses a practical information problem: two funds can hold many of the same companies, but reading monthly portfolio disclosures side by side is laborious. The application converts selected official AMC workbooks into validated holdings keyed by ISIN and calculates weighted overlap, constituent breadth, concentration, effective holdings, month-to-month turnover, and selected large-cap mandate indicators.

The system combines immutable source files, format-specific extraction adapters, source-row and SHA-256 provenance, a deterministic Decimal-based analytics engine, a FastAPI service and a responsive static dashboard. Independent overlap calculations provide arithmetic cross-checks. The application does not use a language model to produce portfolio metrics. Any optional metadata research is separately triggered, restricted to official AMC domains, and requires a cited numeric value before accepting a field.

The frozen evidence window spans July 2025 to June 2026. The source manifest contains 76 archived files totalling 188,908,286 bytes. Complete monthly ISIN-level holdings are available for HDFC Large Cap, SBI Large Cap, Nippon India Large Cap and Tata Large Cap; HDFC Nifty 50 Index is retained as a control. The stated research question originally selected four schemes including ICICI Prudential Large Cap. Its monthly factsheets omit holding ISINs and aggregate some small positions, therefore the system cannot make a complete four-scheme ISIN comparison and does not substitute another fund.

For June 2026, pairwise overlap among the four usable active large-cap peers ranged from 46.26% (HDFC–SBI) to 53.46% (Nippon India–Tata). These values describe shared disclosed equity weight under the minimum-weight formula, not return correlation. Across the 12 observations, mean one-way monthly turnover ranged from approximately 3.81% for HDFC to 6.74% for Tata. The results demonstrate the feasibility of a transparent descriptive workflow; they do not establish portfolio superiority, suitability, predictive performance or accounting-grade completeness.

PRINCIPAL CONCLUSION

FactSheetAI can reproducibly compare available ISIN-level monthly equity disclosures and explain how each figure is calculated. The current source coverage is insufficient to answer the original four-fund question in full; missing ICICI holdings and incomplete reconciliation must remain explicit.

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The report is organised around the supplied index. The analysis chapters describe the application as implemented and use repository-derived source coverage and calculations. Institutional fields on the cover and certificate are intentionally left for the student and institution to complete.

Project-Specific CRAFT Prompt

CRAFT element	Prompt specification
Context	The project is FactSheetAI, an academic software prototype that analyses publicly disclosed monthly equity portfolios of Indian mutual fund schemes using archived official AMC holdings and effective-dated AMFI market-cap lists. Current complete coverage spans HDFC, SBI, Nippon India and Tata Large Cap plus an HDFC Nifty 50 index control, from July 2025 through June 2026. ICICI Prudential Large Cap is a requested scheme but its archived factsheets do not provide complete ISIN-level holdings.
Role	Act as an academic researcher and software-systems analyst with working knowledge of mutual-fund portfolio disclosures, data provenance, deterministic analytics, research methodology and responsible interpretation. Treat source disclosure as evidence, distinguish implementation facts from evaluation and proposals, and avoid investment recommendations.
Action	Explain the research problem and objectives; document the source acquisition, validation, extraction and analytical workflow; interpret overlap, Jaccard breadth, concentration, effective stock count, turnover and effective-date large-cap exposure; assess system architecture and validation evidence; identify missing evidence and limitations; and give prioritised, testable recommendations. Never infer missing holdings or claim the complete four-fund result.
Format	Prepare a formal college project report of approximately 30–35 pages with a title page and front matter, executive summary, this CRAFT prompt, the nine numbered chapters in the supplied index, tables for corpus and measured results, source-linked references, and annexures for formulas, data dictionary, provenance and evaluation checklist. Use consistent headings, page numbers, captions and a professional Word layout.
Tone	Use precise, neutral academic prose. Label computed observations, design choices, interpretations and recommendations distinctly. Define technical terms at first use. State evidence periods, denominators and data gaps beside relevant results. Avoid promotional claims, fabricated citations, unsupported performance claims and financial advice.

1 Introduction to the Study

1.1 Background

Mutual funds pool contributions from investors and invest them in portfolios governed by scheme objectives and regulatory disclosures. An investor who holds multiple schemes may therefore have exposure to the same listed companies through several funds. Such overlap can be intentional, but it can also create less diversification than the number of fund names suggests. A portfolio-level view of common securities can help describe this duplication more clearly than fund labels alone.

Monthly portfolio disclosures are published in heterogeneous formats. A researcher must locate the correct scheme and month, identify the equity section, standardise identifiers, interpret percentages and preserve the source context. Manual comparisons are slow and vulnerable to identity errors. Fund names and tickers are not always stable join keys; a security identifier such as the International Securities Identification Number (ISIN) offers a more precise basis for matching the same security across disclosures.

FactSheetAI turns this workflow into a source-backed computational process. It parses selected official disclosures, validates structured holdings, computes descriptive metrics and presents the results through an API and web dashboard. The system is intentionally framed as an educational research instrument, not an investment decision engine.

1.2 Problem Statement

The central problem is that portfolio disclosures are available as separate files and do not directly show how much equity weight is shared across schemes. Without a common identifier and consistent calculation, a reader may rely on informal name matching or conflate holdings overlap with performance correlation. Both approaches can misstate what the disclosed portfolio evidence supports.

The initial project question concerns overlap among four selected large-cap funds. However, the archived source corpus does not contain complete ISIN-level holdings for ICICI Prudential Large Cap Fund. Its available monthly factsheets omit holding ISINs and aggregate some positions below one per cent. Consequently, the intended complete four-fund comparison cannot be computed from the current evidence. The system instead compares pairs and sets whose source files are available, while retaining the missing scheme in the catalogue as unavailable.

RESEARCH QUESTION

How can a reproducible, auditable software workflow quantify common disclosed equity exposure and portfolio change among source-available Indian large-cap mutual fund schemes, while preserving uncertainty and missing-data boundaries?

1.3 Need and Significance

The project has methodological value because it combines an explicit identity key, deterministic calculations, immutable source records and row-level provenance. It also has educational value: formulas such as weighted overlap and Herfindahl–Hirschman concentration can be connected to actual source rows. The prototype makes a bounded claim about disclosed holdings and makes it possible to inspect how data moves from source files to comparative outputs.

1.4 Conceptual Foundations

Portfolio overlap is defined here as the sum of the smaller reported weights for securities present in both portfolios. It estimates the portion of each fund's NAV represented by shared equity positions under the disclosed weights. It is not a statistical measure of co-movement. Jaccard similarity compares the count of common holdings with the count of all distinct holdings and does not incorporate position size.

Concentration describes how portfolio weight is distributed. Top-N concentration sums the largest N equity weights. The Herfindahl–Hirschman Index (HHI) sums squared fractional weights; its reciprocal is interpreted as an effective number of equally weighted securities. Turnover is estimated from consecutive disclosures as one-half of the sum of absolute weight changes, treating exits and entries as zero in the missing month. These metrics are descriptive and rely on comparability of source-reported portfolio weights.

The project also compares a scheme's holdings against an effective-dated AMFI top-100 company list. This is an operational proxy for large-cap exposure in this prototype. Membership is evaluated using the list in force for the month; it does not classify non-members as mid-cap or small-cap and does not substitute for full regulatory portfolio classification.

2 Objectives of the Study

2.1 Primary Objective

To design and document a reproducible software system that analyses overlap, breadth, concentration and monthly change in source-available Indian mutual-fund equity portfolios using official monthly disclosures and security-level identifiers.

2.2 Specific Objectives

1. Identify official monthly source records and preserve them without modification, alongside source URLs, disclosure months, file sizes and cryptographic hashes.
2. Map requested schemes to exact AMFI Direct Growth plan codes and use ISIN as the security join key.
3. Implement format-aware adapters for available AMC spreadsheets and a strictly validated normalized CSV input contract.
4. Calculate pairwise weighted overlap and Jaccard breadth, concentration measures, effective holdings, portfolio drift and effective-date top-100 exposure.
5. Provide independent arithmetic implementations or identities as cross-checks for overlap calculations.
6. Expose catalog and analysis functionality through a FastAPI JSON interface and a responsive static dashboard.
7. Present provenance and data availability so that each result can be traced and missing values remain unavailable.
8. Assess the current corpus and validation evidence, identify gaps that prevent a complete four-fund conclusion, and recommend a responsible next-stage evaluation plan.

2.3 Research Questions

1. What share of reported equity NAV is common to each source-available pair of schemes in a selected disclosure month?
2. How do constituent breadth, concentration and effective stock count differ across the available active large-cap schemes?
3. How much do reported equity weights change between successive monthly disclosures?
4. How does effective-dated AMFI top-100 membership affect the interpretation of large-cap exposure over time?
5. What data and validation conditions are required before the prototype can answer the original four-scheme comparison reliably?

2.4 Expected Academic Contribution

The contribution is a transparent implementation and evidence trail rather than a novel financial theory. The project makes operational definitions inspectable, ties parsed positions to source rows, and documents where the evidence is incomplete. It demonstrates how software design can make a small financial-data study more reproducible while also making the boundary of a valid conclusion visible.

3 Scope and Limitations

3.1 Study Scope

The empirical window contains the twelve disclosure months from July 2025 through June 2026. The scheme universe with monthly ISIN-level files consists of HDFC Large Cap, SBI Large Cap, Nippon India Large Cap and Tata Large Cap. HDFC Nifty 50 Index Fund is included as a control portfolio for descriptive comparison. The original requested group included ICICI Prudential Large Cap; the scheme remains represented in the registry, but no valid ISIN-level holdings from its factsheet PDFs are loaded into the overlap engine.

The subject matter is equity holdings. Cash, debt, derivatives and other non-equity buckets may be reported differently or may be omitted by the equity-only analytics. The system examines reported weights, not market returns. It includes source ingestion, validation, calculations, API and dashboard functionality and selected Excel report-generation code.

Dimension	Included in present study
Geography / market	Indian mutual-fund scheme disclosures
Observation window	July 2025–June 2026 (12 months)
Primary identity	ISIN; exact AMFI scheme code for plan mapping
Portfolio data	AMC monthly equity holdings and reported NAV weights
Reference universe	AMFI half-yearly top-100 company lists, effective dated
Available schemes	Four active large-cap schemes plus one index control
System layer	Python parsers, deterministic analytics, FastAPI, static dashboard

3.2 Limitations

- The complete originally requested four-fund comparison is unavailable because ICICI Prudential factsheets omit holding ISINs and aggregate some small positions. A peer fund is not a substitute for the missing requested scheme.
- The corpus has 76 files totalling 188,908,286 bytes, below the documented 300 MB genuine-data target. Coverage is concentrated in five scheme series and one year.
- No hand-labelled gold set exists. Parser precision and recall are unknown, and adapter confidence values are declarations rather than measured probabilities.
- NAV and AUM reconciliation is incomplete. Reported equity weights may not sum to total NAV because of cash, debt, derivatives, disclosure conventions or parser scope.
- Official benchmark constituent portfolios are unavailable; active share cannot be established as a valid scheme-versus-benchmark result.
- Sector classifications and some metadata are absent or inconsistent. AUM, expense ratio, category average, manager and cash values are left unavailable unless a traceable source supports them.
- Monthly turnover is calculated from disclosed weight changes. It is a portfolio-change measure and should not be presented as a brokerage-style reported turnover ratio.
- The optional metadata search is user-triggered and depends on API configuration. It is not part of the deterministic calculation pipeline.
- The project has no authentication, database persistence, scheduler, review queue, production monitoring or arbitrary PDF layout extraction.
- The Excel export route requires LibreOffice for recalculation and error scanning; the current documented environment lacks a verified LibreOffice runtime.

3.3 Delimitations

The study intentionally avoids estimating missing holdings, filling gaps by fuzzy company-name matching or fabricating missing market values. It also avoids investment advice, suitability scoring, forecasts and claims that a lower or higher overlap is inherently preferable. These delimitations protect the interpretation from exceeding the data and implementation evidence.

4 Research Methodology (Secondary Data Sources)

4.1 Research Design

This is a descriptive, computational case study based on secondary documentary data. It combines a software engineering evaluation with quantitative analysis of monthly portfolio snapshots. The unit of observation is a scheme-month portfolio disclosure. The primary comparison unit is a pair of scheme-month snapshots from the same disclosure month. No survey, interview or experimental intervention was conducted.

4.2 Data Sources

The primary sources are original AMC monthly portfolio disclosures archived in the project's data/raw directory. The source set contains workbook files for HDFC, SBI, Nippon India and Tata and factsheet PDFs for ICICI. The AMFI NAV file is used to validate scheme codes and exact Direct Growth names. AMFI half-yearly market-cap workbooks define the top-100 list effective for each half of the observation period. The source index stores official URLs and disclosure months; the generated manifest stores SHA-256 hashes and byte counts.

Source group	Files in manifest	Study role
AMC monthly holdings	60	12 months × 5 series (four active peers and one HDFC index control); ICICI PDFs are reference-only
AMFI NAV reference	1	Scheme-name and code mapping
AMFI market-cap lists	3	Effective-dated top-100 reference lists
Total	76	188,908,286 bytes archived

4.3 Sampling and Inclusion Rules

The corpus is a purposive sample of official sources already archived in the project. A scheme-month is analytically eligible when a configured source adapter can extract an unambiguous portfolio for the requested scheme and month with ISIN-bearing equity rows. The analysis window is fixed to twelve consecutive months. For a given pair, both schemes must have eligible holdings for the same month.

ICICI's monthly factsheets are excluded from holding-level analysis because they do not provide complete ISIN-level constituent information. The exclusion is not a claim that the fund lacks holdings; it is a statement about the fields disclosed in the archived source. HDFC Nifty 50 is labelled a control and is not represented as a replacement for a requested active scheme.

4.4 Source Preservation and Provenance

Original source files are byte-preserved. The manifest records relative path, source type, source eligibility, source URL, disclosure month, file size and SHA-256. Parsing produces holding-level provenance fields including scheme code, source filename, source hash, AMC, month, parse timestamp and source-row or source-span references. Results can therefore be reconstructed from source bytes, parser code and the applicable configuration, subject to the same software environment.

The provenance design is central to the methodology: a number is not treated as sufficiently supported merely because it is computable. The reader should be able to identify which disclosure generated it and which row supported a parsed position. If a value cannot be read reliably or a source is absent, the application reports it as unavailable.

4.5 Data Extraction and Validation

Format-specific adapters select holdings within labelled equity sections of the relevant AMC sheets. They validate the requested scheme and disclosure month, parse ISINs and weights, preserve original row spans, and reject ambiguous fund matches or conflicting duplicate security names. Duplicate same-ISIN rows are only aggregated where normalized names agree after source footnote markers are removed. Unsupported weight values are rejected rather than silently coerced.

The normalized CSV fallback requires one scheme-month per file and exact columns for scheme identity, disclosure month, ISIN, security name, percentage weight, asset type and AMC. Pydantic models constrain ISIN shape and weight range and reject duplicate equity identifiers. This contract supports controlled user-supplied normalized data; it is not arbitrary PDF extraction or production-corpus ingestion.

One special documented case occurs in the SBI April 2026 workbook: a marked weight is derived from the same row's disclosed market value and the workbook's grand-total AUM, and the derivation is preserved in the source span. The methodology identifies this as a point requiring source-review sign-off and independent accounting reconciliation.

4.6 Analytical Procedures

Metric	Operational definition	Interpretive boundary
Weighted overlap	Sum of min(wA, wB) across common equity ISINs	Shared reported NAV weight; not correlation
Jaccard breadth	Common ISIN count ÷ union ISIN count	Equal-count measure; ignores position size
Top-N concentration	Sum of N largest equity NAV weights	Concentration within parsed equity holdings
HHI / effective count	Sum (wi/100)^2; reciprocal of HHI	Effective equally weighted stock count
Monthly turnover / change	Half the sum of absolute weight differences across union	Descriptive change between disclosed snapshots
Top-100 exposure	Sum of equity weights in month-effective AMFI top-100 list	Operational large-cap proxy; not complete classification

The overlap calculation is independently reproduced using a pandas merge and a pure-Python mapping; an algebraic identity provides an additional check in the analytics layer. Portfolio figures use decimal arithmetic. Red-flag and health-score thresholds are configurable rules and must be read as prototype diagnostics, not regulatory judgements or investment recommendations.

4.7 Effective-Dated Reference Data

The June 2025 AMFI list is applied to July–December 2025; the December 2025 list is applied to January–June 2026. This prevents a later ranking from being applied retrospectively. A June 2026 effective list exists for the next half-year, but it falls outside the portfolio observation period. Each reference workbook is tracked separately with effective date and source hash.

5 Data Analysis and Interpretation

5.1 Corpus Coverage

The manifest includes 76 source files and 188,908,286 bytes. The study period has 12 monthly records for each of five archived series: four active large-cap peers plus the HDFC Nifty 50 control. Three half-yearly AMFI market-cap workbooks and one AMFI NAV reference are retained. The 12 ICICI PDFs remain reference-only because the extracted disclosures do not support complete ISIN joins.

Scheme series	Eligible months	Mean holdings per snapshot	July 2025 holdings	June 2026 holdings
HDFC Large Cap	12 / 12	46.8	48	44
SBI Large Cap	12 / 12	43.8	37	53
Nippon India Large Cap	12 / 12	67.8	69	64
Tata Large Cap	12 / 12	43.3	42	44
HDFC Nifty 50 Index (control)	12 / 12	50.2	50	50
ICICI Prudential Large Cap	0 / 12 ISIN-level	Not available	Not available	Not available

The average count is based on parsed snapshot holdings for the series. It is not a claim that every AMC's economic portfolio contains only this many assets; excluded or unparsed assets and disclosure conventions remain relevant. The unequal counts, especially Nippon's larger mean, make the breadth metric useful alongside weighted overlap.

5.2 June 2026 Pairwise Overlap

The June 2026 comparison uses the last month in the frozen study window. Weighted overlap is the sum of the lower weight for each common ISIN, measured in percentage points of NAV as reported. The matrix below is symmetric; the diagonal is not shown because self-overlap is not the focus.

Scheme pair	Weighted overlap	Common ISINs	Jaccard
HDFC – SBI	46.26%	16	0.198
HDFC – Nippon India	44.70%	22	0.256
HDFC – Tata	52.00%	19	0.275
SBI – Nippon India	51.07%	24	0.258
SBI – Tata	51.89%	23	0.311
Nippon India – Tata	53.46%	23	0.271

Among the six available active-peer comparisons, overlap ranges from 44.70% to 53.46%. HDFC–SBI is the lowest pair in this month, while Nippon India–Tata is the highest. This describes a substantial shared reported equity allocation for each pair under the chosen formula. The value does not state that 44.70–53.46% of returns will move together, nor does it include a complete four-fund consolidated portfolio with ICICI.

5.3 Control Comparisons and Breadth

Active scheme vs HDFC Nifty 50 control	Weighted overlap	Common ISINs	Jaccard
HDFC Large Cap	53.48%	24	0.343
SBI Large Cap	53.53%	22	0.272
Nippon India Large Cap	60.91%	26	0.295
Tata Large Cap	59.94%	23	0.324

The control comparisons range from 53.48% to 60.91%. The index control contains 50 holdings in June 2026 and serves as a reference portfolio in the corpus. These comparisons should not be read as active share. Active share requires a valid and comparable benchmark portfolio definition, complete weights and appropriate treatment of cash; the project documents official benchmark constituents as unavailable for that purpose.

Jaccard adds a different view. In June 2026 HDFC–SBI share 16 ISINs, corresponding to a Jaccard value of 0.198 despite a weighted overlap of 46.26%. A small set of shared names can carry material weights. Conversely, broader common membership does not necessarily imply a higher weighted overlap if shared positions are small. Reporting both values avoids treating breadth and exposure as interchangeable.

5.4 Longitudinal Concentration and Equity Weight

Scheme	Jul-25 equity weight	Jun-26 equity weight	Jul-25 Top-10	Jun-26 Top-10	Jun-26 HHI	Effective stocks Jun-26
HDFC	95.96%	96.76%	51.17%	51.70%	0.03889	25.71
SBI	95.29%	97.95%	51.55%	48.30%	0.03618	27.64
Nippon India	98.87%	98.87%	44.49%	45.75%	0.03409	29.34
Tata	96.45%	98.04%	46.93%	43.88%	0.03272	30.56
HDFC Nifty 50 control	99.76%	99.69%	56.29%	53.42%	0.04280	23.36

The table reports metrics over parsed equity holdings. Between July 2025 and June 2026, the four active schemes' reported equity sums increase modestly in the archived snapshots. HDFC and SBI Top-10 concentration changes in different directions: HDFC rises from 51.17% to 51.70%, while SBI falls from 51.55% to 48.30%. Nippon India moves from 44.49% to 45.75%, and Tata declines from 46.93% to 43.88%.

5.5 Interpreting HHI and Effective Holdings

HHI squares each security's fractional NAV weight before summing. A larger value indicates greater concentration under this definition. The reciprocal expresses the number of equally weighted holdings that would produce the same HHI. It is an equivalence measure, not an actual count of securities. In June 2026 the active-fund effective-count values range from approximately 25.71 for HDFC to 30.56 for Tata, despite raw holding counts ranging from 44 to 64.

The 50-stock index control has a June HHI of 0.04280 and an effective count of 23.36. This is consistent with the general mathematical property that unequal weights reduce effective count relative to raw count. Comparisons remain sensitive to the NAV-weight denominator and the completeness of extracted positions. HHI should therefore be read beside total parsed equity weight and source coverage, rather than as a standalone score.

5.6 Month-to-Month Portfolio Change

Scheme	Mean monthly one-way change	Maximum monthly change
HDFC Large Cap	3.81%	9.43%
SBI Large Cap	4.88%	8.26%
Nippon India Large Cap	4.61%	7.96%
Tata Large Cap	6.74%	13.69%
HDFC Nifty 50 control	2.60%	5.63%

The measure is half the sum of absolute month-to-month equity weight changes across the union of ISINs, so entries and exits are included. Tata has the highest mean and maximum observed change in this sample; the index control has the lowest. These are changes in disclosed weights, not observed transactions, manager intent or a formally reported turnover ratio. Valuation changes can affect weights even when no trading occurs.

5.7 Large-Cap Reference Exposure

For July 2025, HDFC Large Cap has 90.81% of parsed equity weight in the AMFI top-100 list effective 30 June 2025; 5.15% is outside that list. The project uses an 80% configured floor, so the July observation is above the prototype threshold. For June 2026, HDFC has 77.62% in the top-100 list effective 31 December 2025 and 19.14% outside, placing it below the configured 80% floor in the prototype calculation.

This shift warrants careful interpretation rather than a claim of a confirmed regulatory breach. The method compares reported equity rows with effective-dated top-100 membership; it does not reconstruct the complete NAV, determine all asset classifications or independently establish compliance. Non-members are labelled outside the top-100 set, not automatically classified into another market-cap category. The observation illustrates why effective dating and the denominator should be disclosed beside the result.

5.8 Reproducibility and Cross-Checks

The codebase documents parallel pandas and pure-Python overlap implementations and an identity based on $\min(a,b) = (a+b-|a-b|)/2$. Unit and property-based tests exercise arithmetic invariants on both hand-calculated and fixture-based examples. Ingestion fixtures include valid, empty, malformed-header, mixed-month, concentration and style cases. Synthetic fixtures are kept under tests/fixtures and excluded from the production raw-source catalogue.

These checks increase confidence that the formulas are implemented consistently, but they do not establish source extraction accuracy. A parser may consistently produce the wrong weight if the source layout was misread. No labelled gold set, independent source reconciliation across all schemes, or completed NAV/AUM reconciliation exists. Arithmetic validation and data validity are separate evaluation dimensions.

6 Findings and Discussion

6.1 Principal Findings

1. A provenance-first pipeline is implemented for four active large-cap series and one index control, with 12 monthly files per series in the study window.

2. In June 2026, weighted overlap among active peer pairs lies between 44.70% and 53.46%; common constituents range from 16 to 24. This supports the descriptive observation that shared positions can account for a meaningful share of reported equity weights.

3. Jaccard and weighted overlap rank portfolios differently because one measures constituent count and the other weights shared positions. The metrics should be read together.

4. Top-10 concentration and HHI vary across schemes. Tata has the highest June effective stock count among active peers in this corpus, while HDFC has the lowest. This is a mathematical concentration comparison, not a ranking of fund quality.

5. Observed one-way monthly portfolio change is highest for Tata and lowest for the index control in the one-year sample. It describes changing weights and not necessarily transactions.

6. The effective-dated AMFI list materially affects the interpretation of large-cap exposure. The HDFC example is above the configured floor at the start and below it at the end of the sampled period.

7. The intended complete four-scheme result is unavailable. Since ICICI lacks complete ISIN-level holdings, no complete conclusion about the original requested set can be drawn.

6.2 Discussion of System Design

The project's strongest design choice is separation of source acquisition, parsing, normalized snapshots, analytics and presentation. Immutable raw files and hashes establish a stable evidence base. Per-AMC adapters accommodate differences in workbook formats without encouraging fuzzy matching. The API returns source-related information, and the interface can present month selection, overlap views and portfolio scenarios for available schemes.

Deterministic calculation is appropriate for metrics with explicit formulas. Decimal arithmetic avoids avoidable floating-point variation in many portfolio operations, while the independent implementations reduce the risk of a single-code-path arithmetic defect. The absence of a language model in the calculation and commentary path limits unsupported generated conclusions in this release. The optional web-search metadata function has a separate role and can leave unsupported fields unavailable.

These design controls do not remove the need for human review. Source layouts can change; a parser may miss rows or select the wrong section. The SBI April weight derivation merits explicit review. A hash proves identity of archived bytes, not correctness of issuer disclosure. A source-backed workflow is more auditable than an untraceable output, but traceability is not equivalent to source completeness.

6.3 Discussion of Evidence Adequacy

Evidence is adequate to demonstrate the mechanics of pairwise comparison for the available scheme-months and to provide a reproducible one-year descriptive analysis. It is not adequate to answer the originally framed complete four-fund question, measure parser accuracy, reconcile every equity total to total NAV or establish financial performance implications. This distinction should guide how the project is presented in a viva or submitted as an academic report.

6.4 Validity Considerations

Validity dimension	Current evidence	Implication
Construct validity	Explicit overlap, breadth, concentration and change formulas	Metrics describe their defined constructs; overlap is not correlation
Internal / computational validity	Deterministic engine, dual overlap paths, invariant tests	Arithmetic errors are partly controlled; parser errors remain possible
Source validity	Official source archive, URL, hash, month and row spans	Files are traceable; completeness and disclosure accuracy are not proven
External validity	Five scheme series, one year, Indian large-cap category	Results should not be generalised to other categories, markets or periods
Criterion validity	No hand-labelled gold set or third-party baseline	Precision/recall and accounting-grade correctness remain unknown

6.5 Ethical and Responsible Use

The system is intended for educational analysis. It should not be used to recommend a fund, make a suitability decision, trade securities, or produce regulatory reports. Portfolio evidence may be incomplete, delayed or formatted differently across sources. Any public presentation should show the observation month and source limitations. User-uploaded normalized CSV previews are temporary and should be reviewed before interpretation.

The application's data governance notes appropriately reject fabricated values and emphasize source lineage. Future AI-assisted text generation, if added, should cite retrieved source spans, distinguish computed values from retrieved facts, reject unsupported questions and validate every numeric token. The current project does not implement a retrieval-grounded conversational layer; it should not be described as an AI fact-checking or recommendation system.

7 Conclusion and Suggestions

7.1 Conclusion

FactSheetAI demonstrates a practical way to turn selected public monthly portfolio disclosures into reproducible equity-overlap and drift analysis. Its core workflow uses exact scheme mapping, ISIN identity, source-preserving adapters, validated normalized holdings, deterministic calculations, a JSON API and an interactive dashboard. For available source pairs, the project can report weighted overlap, Jaccard breadth, concentration, effective holdings and monthly changes, while linking results to archived source evidence.

The one-year corpus supports bounded descriptive findings. June 2026 active-peer overlap ranges from 44.70% to 53.46%; the HHI-derived effective stock counts span roughly 25.71 to 30.56 among active peers. Mean monthly one-way change spans approximately 3.81% to 6.74%. Those values illustrate the application's analytical capability but do not establish risk-adjusted performance or investor suitability.

The key limitation is decisive: ICICI Prudential Large Cap does not have complete ISIN-level holdings in the archived source set. The project therefore cannot answer the original complete four-fund comparison. Missing observations, absent gold-standard parser evaluation and incomplete NAV/AUM reconciliation also limit the strength of any broader claim. The appropriate conclusion is that the system is a useful research prototype with transparent boundaries, not a production-grade financial decision tool.

7.2 Suggestions for Improvement

Priority	Recommendation	Evidence of completion
1	Acquire complete official ISIN-bearing ICICI monthly portfolio files and verify the same 12-month window. Keep missing months unavailable until found.	Source URL, original file, hash, disclosure month and parser evidence for each month
2	Build a hand-labelled gold set spanning at least five schemes, source formats and months; independently review every parsed row and weight.	Per-field precision/recall and adjudicated mismatch log
3	Reconcile extracted equity, cash, debt and derivative totals to source portfolio totals and AUM. Review the SBI April derivation separately.	Signed reconciliation workbook with exceptions and source spans
4	Validate effective-dated AMFI mappings, source URLs and amendment history before extending the time window.	Versioned reference-data review record
5	Add source-specific parser fixtures for format changes and fail closed when headers, section boundaries or dates are ambiguous.	Automated regression evidence for every adapter
6	Complete secure deployment controls including authentication, rate limiting, persistence policy and audit events before multi-user use.	Threat model, access-control test and deployment checklist
7	Test Excel export recalculation in a host with verified LibreOffice and compare formulas to API calculations.	Recalculated workbook and formula-parity record
8	If natural-language retrieval is introduced, evaluate citation support and numeric fidelity against a labelled question set before enabling it.	Grounded-answer evaluation set and acceptance thresholds

7.3 Future Research

A longer, complete panel could examine whether overlap persistence differs by scheme pair and how that relates to portfolio concentration or benchmark composition. Such work would require complete source coverage and a pre-registered analytic question. Additional useful studies could compare reported and reconstructed turnover, test sensitivity to incomplete equity weights, quantify the effect of small aggregated positions, and assess parser accuracy across format changes. The present repository provides a foundation for those questions but does not answer them.

8 Bibliography / References

8.1 Primary and Project Sources

1. Association of Mutual Funds in India (AMFI). NAVAll.txt scheme reference archive, retrieved 19 October 2026. Project copy: data/raw/amfi/nav.txt. Used to resolve scheme codes and exact plan names.

2. Association of Mutual Funds in India (AMFI). Half-yearly average market capitalisation workbooks: effective 30 June 2025, 31 December 2025 and 30 June 2026. Project copies: data/raw/amfi/amfi_caps_2025-06.xlsx, amfi_caps_2025.xlsx and amfi_caps_2026.xlsx.

3. HDFC Mutual Fund. Monthly portfolio disclosures for HDFC Large Cap Fund and HDFC Nifty 50 Index Fund, July 2025–June 2026. Archived in data/raw/amc/hdfc/; source URLs and file hashes are recorded in data/raw/manifest.json.

4. SBI Mutual Fund. Monthly portfolio disclosures for SBI Large Cap Fund, July 2025–June 2026. Archived in data/raw/amc/sbi/; source URLs and file hashes are recorded in the project manifest.

5. Nippon India Mutual Fund. Monthly portfolio disclosures for Nippon India Large Cap Fund, July 2025–June 2026. Archived in data/raw/amc/nippon-india/; source URLs and hashes are recorded in the project manifest.

6. Tata Mutual Fund. Monthly portfolio disclosures for Tata Large Cap Fund, July 2025–June 2026. Archived in data/raw/amc/tata/; source URLs and hashes are recorded in the project manifest.

7. ICICI Prudential Mutual Fund. Monthly factsheets for the observation window. Archived reference PDFs are not eligible for ISIN-level analysis because the factsheet does not provide complete ISINs and aggregates some small equity positions.

8. FactSheetAI project documentation. README.md; METHODOLOGY.md; GOVERNANCE.md; TRACEABILITY.md; PROGRESS.md; VIVA.md; config/market_cap_lists.json; config/mandates.json; config/analytics_thresholds.json. Accessed 20 October 2026.

9. FactSheetAI project source code. factsheetai/ingestion.py; models.py; analytics.py; catalog.py; amfi.py; api.py; excel_report.py; report_metadata.py and static/index.html. Accessed 20 October 2026.

8.2 Technical References

10. Python Software Foundation. Python documentation. https://docs.python.org/3/ (accessed 20 October 2026).

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12. Pydantic. Pydantic documentation. https://docs.pydantic.dev/ (accessed 20 October 2026).

13. pandas development team. pandas documentation. https://pandas.pydata.org/docs/ (accessed 20 October 2026).

14. openpyxl contributors. openpyxl documentation. https://openpyxl.readthedocs.io/ (accessed 20 October 2026).

15. Herfindahl, O. C. (1950). Concentration in the steel industry. Unpublished doctoral dissertation, Columbia University. HHI is used here only as a standard concentration measure; portfolio interpretation is defined operationally in the project methodology.

16. Jaccard, P. (1912). The distribution of the flora in the alpine zone. New Phytologist, 11(2), 37–50. The coefficient is used as a count-based similarity measure over ISIN sets.

17. Association of Mutual Funds in India. Industry and investor information portal. https://www.amfiindia.com/ (accessed 20 October 2026).

Repository source index and generated manifest provide exact source URLs for each archived AMC disclosure. The workbook files are primary data records; in a printed submission, an accompanying electronic archive or institution-approved repository should be used for complete source-level traceability.

9 Annexures

Annexure A Formula Sheet

Measure	Formula / convention
Weighted overlap	O(A,B) = Σ(i ∈ A∩B) min(wAi, wBi). Reported as percentage points of NAV.
Equivalent overlap identity	min(a,b) = (a + b − |a − b|) / 2 for each common ISIN.
Jaccard similarity	J(A,B) = |A∩B| / |A∪B|.
Top-N concentration	TopN = Σ of the N largest disclosed equity NAV weights.
HHI	HHI = Σ(wi / 100)^2 for NAV-percentage weights.
Effective stock count	N_eff = 1 / HHI, interpreted as an equivalent equal-weight count.
One-way monthly change	0.5 × Σ over union of ISINs |w_current − w_previous|.
Large-cap reference exposure	Σ weights for equity ISINs in the top-100 set effective in the disclosure month.
Weighted holding exposure	For an allocation fraction ai, combined exposure to ISIN i is Σ(ai × wi).

Annexure B Data Dictionary

Field	Meaning	Control / caveat
scheme_id	AMFI scheme code for selected plan	Exact registry mapping; plan convention is Direct Growth
disclosure_month	Portfolio month represented by source	Must match file-reported month
isin	Security identifier	Uppercase 12-character alphanumeric; primary join key
security_name	Source-reported security name	Descriptive label; not used as a fuzzy match key
weight_pct	Reported percentage of NAV	Decimal percent; validated between 0 and 100
asset_type	Asset class label	Equity-only metrics exclude other types
sha256	Hash of source bytes	Supports file identity, not source correctness
source_row / span	Original workbook row or source location	Audit path to the parsed holding
effective_date	Start date for AMFI list applicability	Month must use list in force; no retrospective update

Annexure C System Architecture and Traceability

Stage	Project component	Evidence output
1. Source archive	data/raw/amc, data/raw/amfi	Original files, source_index.json and manifest.json
2. Configuration	config/schemes.json, market_cap_lists.json, mandates.json	Scheme mapping and effective-dated rules
3. Ingestion	factsheetai/ingestion.py	Validated holdings with source row spans and hashes
4. Analytics	factsheetai/analytics.py	Overlap, concentration, drift, turnover and flags
5. Service	factsheetai/api.py	Catalogue, pair analysis, multi-fund endpoints
6. Interface	factsheetai/static/index.html	Month selection, heatmap, holdings and what-if views
7. Test evidence	tests/ and tests/fixtures/	Parser, API and arithmetic behaviour; no gold-set evaluation

The Excel report generator is also integrated in factsheetai/excel_report.py. Its output depends on sourced metadata and an available LibreOffice recalculation step. The repository documentation records that this dependency was not verified in the current environment, so export recalculation and formula-error scanning remain an open validation item.

Annexure D Evaluation Checklist

- Confirm original file URL, disclosure month, scheme identity and SHA-256 before parsing.
- Check sheet names, section boundaries, column headers, ISIN format, duplicate rows and percentage scale.
- Compare parsed row count and equity total with the source disclosure; record all exclusions and derivations.
- Review the source row or span for every exceptional value and each sampled calculation.
- Recalculate overlap using an independent method and confirm symmetric pair results.
- Use the AMFI list that is effective for the portfolio month and preserve its date and hash.
- Label missing source files as unavailable; do not interpolate or substitute another scheme.
- Display the disclosure month, formula definition and scope limitations with any exported figure.
- Before production use, complete a labelled parser evaluation, NAV/AUM reconciliation, security review, access control and operational monitoring.

Annexure E Abbreviations

Abbreviation	Expansion
AMC	Asset Management Company
AMFI	Association of Mutual Funds in India
API	Application Programming Interface
HHI	Herfindahl–Hirschman Index
ISIN	International Securities Identification Number
NAV	Net Asset Value
CSV	Comma-Separated Values
SHA-256	Secure Hash Algorithm, 256-bit